ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** water	Splitting of water.
2	**(c)** herbivores such as the cow	Cellulose needs long digestion.
3	**(d)** pulmonary artery	Only artery with deoxygenated blood.
4	**(b)** oxygen	Oxygen is a product.
5	**(d)** cytoplasm	Cytoplasm.
6	**(b)** 120/80 mm Hg	Standard.
7	**(a)** guard cells take up water and swell	Turgid guard cells open pore.
8	**(d)** much more energy per molecule of glucose	Complete breakdown in mitochondria gives more energy.
9	**(d)** xylem	Xylem.
10	**(a)** no blue-black colour	No starch is made in dark.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No light, no photosynthesis (1); Iodine test gives no blue-black colour (1).	1+1
12	Cellulose digestion takes longer (1); Meat is easier to digest (1).	1+1
13	Carries digested fat from intestine (1); Drains excess fluid from tissues (1).	1+1
14	More light energy is absorbed (1); Then other factors like CO_2 limit it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Large flat surface to capture light (1); Stomata for gas exchange (1); Chloroplasts rich in chlorophyll in mesophyll (1).	1+1+1
16	Mixing reduces the oxygen supplied to the body (1); Birds and mammals need more energy (1); Four chambers prevent mixing (1).	1+1+1
17	Waste like urea accumulates (1); Dialysis removes waste (1); Kidney transplant is the long-term cure (1).	1+1+1
	OR: Mucus coats the wall (1); It protects against HCl (1); Acid helps pepsin work (1).	1+1+1
18	Heart valves prevent backflow (1); Vein valves keep blood moving towards heart (1); Blood flows one way (1).	1+1+1
	OR: Distances are long (1); Diffusion is slow (1); So respiratory pigments in blood are used (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Destarch the plant in dark (1); Cover part of a leaf with black paper (1); Expose to sunlight (1); Iodine test: covered part no blue-black (1); Conclusion: light is necessary (1).	1+1+1+1+1
	OR: Herbivores have long intestine (1); Carnivores short intestine (1); Cellulose digestion in herbivores (1); Caecum is large in herbivores (1); Carnivores digest meat easily (1).	1+1+1+1+1
20	Mouth: amylase gives sugar (1); Stomach: pepsin gives peptides (1); HCl provides acid (1); Small intestine: trypsin, lipase, bile (1); End products absorbed (1).	1+1+1+1+1
	OR: Glycolysis in cytoplasm gives pyruvate (1); With oxygen: pyruvate to CO_2 + water in mitochondria (1); Without oxygen: ethanol or lactic acid (1); Aerobic gives more energy (1); Anaerobic in yeast and muscle (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that plants respire only at night and photosynthesise only by day; plants respire all the time and photosynthesise in the presence of light.
- Mixing up xylem and phloem: xylem carries water and minerals (one-way, upwards); phloem carries food (both directions).
- Saying pulmonary artery carries oxygenated blood; the pulmonary **artery** carries deoxygenated blood to the lungs and the pulmonary **vein** carries oxygenated blood back.
- Confusing breathing with respiration; breathing is gas exchange, respiration is the breakdown of glucose in cells to release energy.
- Stating that pyruvate is formed in the mitochondria; glucose is broken down to pyruvate in the cytoplasm.

- Believing the kidney makes urine only by filtration; selective reabsorption and the amount of water excreted are equally important steps.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** lactic acid accumulation	Anaerobic respiration.
2	**(d)** transpiration pull	Pull on water column.
3	**(a)** lose water and become flaccid	In the dark the guard cells lose water and the pore closes.
4	**(d)** cytoplasm	Glycolysis.
5	**(d)** osmosis	Water moves along the concentration gradient.
6	**(b)** trypsin and lipase	Enzymes for proteins and fats.
7	**(d)** flattens	Lung volume increases.
8	**(d)** increase in temperature	More evaporation.
9	**(b)** mucus	Mucus.
10	**(a)** gills	Gills.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Inhalation: diaphragm flattens, ribs lift, air enters (1); Exhalation: diaphragm rises, air leaves (1).	1+1
12	Blood flows at high pressure (1); Thick elastic walls withstand it (1).	1+1
13	Emulsifies fats into small globules (1); Increases surface area for lipase (1).	1+1
14	Dissolved oxygen in water is low (1); More frequent breathing needed (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Increase surface area (1); Rich in blood capillaries (1); Absorb digested food to blood (1).	1+1+1
16	Artery: thick elastic wall (1); Vein: thin wall with valves (1); Capillary: one cell thick for exchange (1).	1+1+1
17	Dissolved in blood plasma (1); Carried to lungs (1); Exhaled through alveoli (1).	1+1+1
	OR: Glucose is partly broken down (1); Pyruvate not completely oxidised (1); Much less ATP is produced than in aerobic respiration (1).	1+1+1
18	Absorbed by root hairs through osmosis (1); Rises through xylem (1); Transpiration pull from leaves (1).	1+1+1
	OR: Plants lack special organs (1); Gas exchange by stomata and lenticels (1); Each part respires on its own (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Alveolar air has more oxygen (1); Oxygen diffuses to blood (1); Haemoglobin carries it (1); Released in tissues (1); CO_2 moves the other way (1).	1+1+1+1+1
	OR: Plasma carries food and waste (1); RBC carry oxygen (1); WBC fight germs (1); Platelets clot blood (1); Lymph carries fats (1).	1+1+1+1+1
20	Right side: deoxygenated blood to lungs (1); Lungs: oxygenated blood to left atrium (1); Left ventricle to body (1); Body to right atrium (1); Advantage: efficient oxygen supply (1).	1+1+1+1+1
	OR: Plants: leaves, gums, resins (1); Humans: kidneys, urine (1); Plants use less energy (1); Plants store waste (1); Humans excrete urea (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that plants respire only at night and photosynthesise only by day; plants respire all the time and photosynthesise in the presence of light.
- Mixing up xylem and phloem: xylem carries water and minerals (one-way, upwards); phloem carries food (both directions).
- Saying pulmonary artery carries oxygenated blood; the pulmonary **artery** carries deoxygenated blood to the lungs and the pulmonary **vein** carries oxygenated blood back.
- Confusing breathing with respiration; breathing is gas exchange, respiration is the breakdown of glucose in cells to release energy.
- Stating that pyruvate is formed in the mitochondria; glucose is broken down to pyruvate in the cytoplasm.

- Believing the kidney makes urine only by filtration; selective reabsorption and the amount of water excreted are equally important steps.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** tubule	Selective reabsorption.
2	**(c)** make the medium suitable for pancreatic enzymes	Neutralises acid from the stomach.
3	**(b)** collapsing	Keep passage open.
4	**(a)** urea, water and salts	Waste products.
5	**(b)** bile salts	Emulsification.
6	**(a)** left atrium and left ventricle	From lungs through left side.
7	**(d)** chlorophyll	Chlorophyll is functional pigment.
8	**(b)** villi	Villi.
9	**(a)** pumping blood to the whole body	High pressure needed.
10	**(c)** blood clotting	Clotting.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Root pressure: pushes water up at night (1); Transpiration pull: main force in the day (1).	1+1
12	Ingestion: taking food in (1); Egestion: removal of undigested food (1).	1+1
13	Systolic when ventricle contracts (1); Diastolic when it relaxes (1).	1+1
14	Uses energy from ATP (1); Sucrose loaded into phloem tissue (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Breathing rate increases (1); More blood flows to the muscles (1); Haemoglobin carries oxygen (1).	1+1+1
16	Transpiration: loss of water vapour (1); Translocation: movement of food (1); Xylem vs phloem (1).	1+1+1
17	Light: guard cells take water, swell, open (1); Dark: lose water, shrink, close (1); Regulates water loss (1).	1+1+1
	OR: Humans: CO_2 + water (1); Yeast: ethanol + CO_2 (1); Place: mitochondria vs cytoplasm (1).	1+1+1
18	Glomerulus: high pressure filtration (1); Tubule: selective reabsorption (1); Capillaries surround the tubule (1).	1+1+1
	OR: Carbohydrates: glucose (1); Proteins: amino acids (1); Fats: fatty acids and glycerol (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Xylem carries water (1); Pull created by transpiration (1); Increased by heat and wind (1); Decreased by humidity (1); Stomata regulate it (1).	1+1+1+1+1
	OR: Amoeba engulfs with pseudopodia (1); Food vacuole forms (1); Digestion within (1); Paramecium uses cilia (1); Waste expelled by rupturing (1).	1+1+1+1+1
20	Filtration (1); Reabsorption (1); Tubular secretion/collection (1); Water content decides the volume (1); Controlled by hormones (1).	1+1+1+1+1
	OR: Breathing brings oxygen (1); Haemoglobin carries it (1); Heart pumps blood (1); Cells oxidise glucose (1); ATP is produced (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that plants respire only at night and photosynthesise only by day; plants respire all the time and photosynthesise in the presence of light.
- Mixing up xylem and phloem: xylem carries water and minerals (one-way, upwards); phloem carries food (both directions).
- Saying pulmonary artery carries oxygenated blood; the pulmonary **artery** carries deoxygenated blood to the lungs and the pulmonary **vein** carries oxygenated blood back.
- Confusing breathing with respiration; breathing is gas exchange, respiration is the breakdown of glucose in cells to release energy.
- Stating that pyruvate is formed in the mitochondria; glucose is broken down to pyruvate in the cytoplasm.

- Believing the kidney makes urine only by filtration; selective reabsorption and the amount of water excreted are equally important steps.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND